MESNA

With ifosfamide/high-dose cyclophosphamide

Prevents haemorrhagic cystitis

CHEMICAL NAME Sodium 2-sulfanylethanesulfonate
CLASS Uroprotective agent (thiol)
TRADE NAMES Uromitexan · Mesnex · Mesna injection

THERAPEUTIC USES / INDICATIONS

- Prevention of haemorrhagic cystitis caused by ifosfamide and high-dose cyclophosphamide.

ADMINISTRATION

- **IV** bolus or infusion, or **PO** (tablets or IV solution diluted in juice/cola).
- Timed with chemotherapy: before, during and after (protocol-specific).
- Oral bioavailability ~50% — oral dose is double the IV dose.

DOSAGES

- **IV regimen:** 20% of ifosfamide/cyclophosphamide dose (w/w) at 0, 4 and 8 h (total 60%).
- **Continuous infusion:** equal to ifosfamide dose (100%), continued 8–12 h after.
- **Oral:** 40% of oxazaphosphorine dose at 2 and 6 h (after initial IV dose).

PHARMACOKINETICS

- Rapidly oxidised to dimesna in blood (inactive), reduced back to mesna in kidney.
- Excreted in urine. **t½:** mesna ~22 min, dimesna ~70 min.

PHARMACODYNAMICS

- Free thiol in urine binds acrolein (toxic metabolite of ifosfamide/cyclophosphamide) → inactive compound → protects bladder urothelium.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, unpleasant taste (oral).
- Headache, fatigue.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- Hypersensitivity, anaphylaxis (more in autoimmune disease).
- False-positive urine ketone tests.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to mesna or thiols.

KEY INTERACTIONS

- None clinically significant (does not affect chemotherapy efficacy).

NURSING MANAGEMENT

- Give doses at exact times relative to chemotherapy.
- Maintain hydration; urine dipstick for blood before and after each dose; strict fluid balance.
- Encourage frequent voiding.

PATIENT EDUCATION

- Take oral doses on time; mixing with juice improves taste.
- Drink plenty of fluids and empty bladder often.
- Report blood in urine or pain on passing urine.

KEY POINTS

- Uroprotectant; binds acrolein in bladder.
- IV 20% of dose at 0, 4, 8 h.
- Given with ifosfamide/high-dose cyclophosphamide.
- False-positive urine ketones.